plant disease

Editor-in-Chief: Alison E. Robertson
Published by The American Phytopathological Society

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November 2016, Volume 100, Number 11
Page 2333

http://dx.doi.org/10.1094/PDIS-05-16-0731-PDN
DISEASE NOTES

Severe Outbreak of Black Rot in Apple Fruit cv. Fuji Caused by *Diplodia seriata* During Pre-Harvest in Maule Region, Chile

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ABSTRACT

Chile leads the export of fresh apple (*Malus* × *domestica*) in the southern hemisphere, with a cultivated area of 37,207 ha. The Maule region (35°25′ S) is the largest production region of the Chilean apple industry, where more than 60% of Chilean commercial apples are planted (www.odepa.cl, 2015). During the last two seasons (2014 through 2016), unusual fruit decays at harvest were observed on 'Fuji' apples at eight orchards in the Maule region. The incidence of symptoms ranged 0.5 to 15% of harvest fruit. The fruit decay symptoms consisted of light to dark-brown, irregularly shaped lesions with initially firm textures. As the disease progressed, fruit rotted completely. To isolate the causal agent, small fragments (2 to 3 mm in length) of tissue

from the lesion margin were placed on PDA modified with Igepal (Díaz and Latorre 2014). Plates were incubated at 22°C for 5 days, and then hyphal tips were subcultured on PDA to obtain pure cultures. Colonies (*n* = 132 isolates) on PDA produced dense white aerial mycelium, becoming dark gray with age. On the underside, colonies were olive green to dark gray to black. After 15 days in PDA, multiple dark-brown mature pycnidia were observed. Aseptate conidia were produced were initially hyaline but become brown to dark brown, rounded at both ends, with mean dimensions of 19 to 25 × 10 to 15 µm (*n* = 40) and a L/W = 1.8. Based on morphological and cultural characteristics, this fungus was identified as *Diplodia seriata* De Not (Phillips et al. 2007; Sutton 2014). To confirm its identity, three loci (ITS, β-tubulin, and EF1-α) were used for further molecular identification, using primers ITS4/ITS5, Bt2a/Bt2b, and EF1-728F/EF1-986R. The sequences of three isolates (Mz-F7, Mz-F9, and Mz-F10) were deposited in the GenBank (accessions nos. KU942431 to KU942433, KU976447 to KU976449, and KU951891 to KU951893 for ITS, β-tubulin, and EF1-α, respectively). BLAST searches indicated 100% identity with *D. seriata* (strain CBS 112555 ex-type). A pathogenicity test was conducted with isolate Mz-F7 on mature Fuji apples. Twenty apple fruit cv. Fuji were surface disinfested with 75% ethanol for 30 s, a 5 mm diameter core was removed with a sterile cork borer. The apples were inoculated with mycelial plugs (5 mm) from 7-day-old cultures of the putative pathogens. An equal number of apples injured but treated with sterile agar plugs were used as negative controls. The apples were placed into a commercial packing box at 20°C for 10 days. The experiments were conducted twice. All inoculated apples developed rot symptoms identical to those described in the original outbreak. No symptoms were observed on the negative controls. Koch's postulates were fulfilled by 100% reisolating *D. seriata* from the inoculated apples. Previously, *B. dothidea*was reported causing Botryosphaeria canker in apple trees in Chile (Latorre and Toledo 1984), but other members of Botryosphaeriaceae had not been reporter. Further studies are needed to better understand the impact of this severe outbreak of *D. seriata* to apple production in Chile.

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